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High capacity power storage system for electric hydraulic fracturing

Abstract

A system for powering electric hydraulic fracturing equipment, the system including a power storage system and electric powered hydraulic fracturing equipment in selective electrical communication with the power storage system. The system further includes at least one circuit breaker between the power storage system and the electric powered hydraulic fracturing equipment, the circuit breaker configured to facilitate or prevent electrical communication between the power storage system and the electric powered hydraulic fracturing equipment.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 16/943,727 filed Jul. 30, 2020, titled HIGH CAPACITY POWER STORAGE SYSTEM FOR ELECTRIC HYDRAULIC FRACTURING, now U.S. Pat. No. 11,542,786, issued Jan. 3, 2023, which claims priority to and the benefit of U.S. Provisional Application No. 62/881,174, filed Aug. 1, 2019, titled HIGH CAPACITY POWER STORAGE SYSTEM FOR ELECTRIC HYDRAULIC FRACTURING, the full disclosures of which are hereby incorporated herein by reference in their entirety for all intents and purposes.

BACKGROUND

1. Field of Invention

(1) This invention relates in general to equipment used hydraulic fracturing operations, and in particular, to electricity storage at a hydraulic fracturing site.

1. Description of the Prior Art

(2) Hydraulic Fracturing is a technique used to stimulate production from some hydrocarbon producing wells. The technique involves injecting hydraulic fracturing fluid into a wellbore at a pressure sufficient to generate fissures in the formation surrounding the wellbore. Hydrocarbons can then flow through the fissures to a production bore. The hydraulic fracturing fluid is typically injected into the wellbore using hydraulic fracturing pumps, which can be powered, in some cases, by electric motors. The electric motors can in turn be powered by generators.

(3) Preserving and extending the life and durability of power generators at an electric hydraulic fracturing site is a priority. This objective, however, can be undermined by overloading power generation equipment. Such overloading reduces the life span of the equipment, and can also create a hazardous environment at a wellsite due to malfunctions and overheating in close proximity with other hydraulic fracturing equipment.

(4) The fast response electricity storage system of the present technology is one viable option to assisting in power distribution, in particular at times when power generation equipment is overloaded. Not only does such a system provide a rapid and effective way to supply power when demand is high, but it also possesses other features that help provide continuous reliable power to hydraulic fracturing equipment.

SUMMARY

(5) One embodiment of the present technology provides a hydraulic fracturing power system, including a power source, a power storage system, and electric powered hydraulic fracturing equipment in selective electrical communication with the power source, the power storage system, or both. The system further includes at least one circuit breaker between the power source, the power storage system, or both, and the electric powered hydraulic fracturing equipment, the circuit breaker having an open position that opens an electric circuit between the electric powered hydraulic fracturing equipment and the power source, the power storage system, or both, and a closed position that closes the electric circuit.

(6) In some embodiments, the power storage system can be at least one solid state battery selected from the group consisting of electrochemical capacitors, lithium ion batteries, nickel-cadmium batteries, and sodium sulfur batteries. Alternatively, the power storage system can be at least one flow battery selected from the group consisting of redox batteries, iron-chromium batteries, vanadium redox batteries, and zinc-bromine batteries. The at least one battery can be rechargeable.

(7) In certain embodiments, the at least one circuit breaker can include a first circuit breaker and a second circuit breaker, the first circuit breaker electrically connected to the power source, and the second circuit breaker electrically connected to the power storage system. Each of the first circuit breaker and the second circuit breaker can be electrically connected to the electric powered

hydraulic fracturing equipment via a common bus. Alternatively, the at least one circuit breaker can be a first circuit breaker, and both the power source and the power storage system can be electrically connected to the first circuit breaker.

(8) In some embodiments, at least one of the power source and the power storage system can be electrically connected to the at least one circuit break via a power line. In addition, the power storage system can be mounted on a trailer. Furthermore, the at least one circuit breaker can be substantially enclosed in a switchgear housing.

(9) Another embodiment of the present technology provides a system for powering electric hydraulic fracturing equipment, the system including a power storage system, electric powered hydraulic fracturing equipment in selective electrical communication with the power storage system, and at least one circuit breaker between the power storage system and the electric powered hydraulic fracturing equipment, the circuit breaker configured to facilitate or prevent electrical communication between the power storage system and the electric powered hydraulic fracturing equipment.

(10) In certain embodiments, the power storage system can be at least one solid state battery selected from the group consisting of electrochemical capacitors, lithium ion batteries, nickel-cadmium batteries, and sodium sulfur batteries. Alternatively, the power storage system can be at least one flow battery selected from the group consisting of redox batteries, iron-chromium batteries, vanadium redox batteries, and zinc-bromine batteries.

(11) In addition, certain embodiments of the technology can also include a power source. In such embodiments, the at least one circuit breaker can include a first circuit breaker and a second circuit breaker, the first circuit breaker electrically connected to the power source, and the second circuit breaker electrically connected to the power storage system. Alternatively, the at least one circuit breaker can be a first circuit breaker, and wherein both the power source and the power storage system are electrically connected to the first circuit breaker.

(12) Some embodiments can include a power source, wherein at least one of the power source and the power storage system are electrically connected to the at least one circuit breaker via a power line, and wherein the at least one circuit breaker is substantially enclosed in a switchgear housing. Furthermore, the power source can be rechargeable. Alternatively, the power source can be electrically connected to the at least one circuit breaker via a power line, and the power storage system can be located adjacent the switchgear housing and electrically coupled directly to the switchgear without a power line.

(13) Additionally, yet another embodiment can include software in communication with the power storage system, the software configured to monitor the state of the power storage system and to integrate control of the power storage system with other features of the system for powering electric hydraulic fracturing equipment.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present technology will be better understood on reading the following detailed description of non-limiting embodiments thereof, and on examining the accompanying drawings, in which:

(2) FIG. 1A is a schematic diagram of a hydraulic fracturing power system according to an embodiment of the present technology;

(3) FIG. 1B is a schematic diagram of a power storage system as used in the embodiment of the hydraulic fracturing power system of FIG. 1A;

(4) FIG. 2A is a schematic diagram of a hydraulic fracturing power system according to an alternate embodiment of the present technology;

(5) FIG. 2B is a schematic diagram of a power storage system as used in the embodiment of the

hydraulic fracturing power system of FIG. 2A;

(6) FIG. 3A is a schematic diagram of a hydraulic fracturing power system according to another alternate embodiment of the present technology;

(7) FIG. 3B is a schematic diagram of an alternate embodiment of the hydraulic fracturing power system of FIG. 3A;

(8) FIG. 4A is a schematic diagram of a hydraulic fracturing power system according to yet another alternate embodiment of the present technology; and

(9) FIG. 4B is a schematic diagram of an alternate embodiment of the hydraulic fracturing power system of FIG. 4A.

DETAILED DESCRIPTION OF THE INVENTION

(10) The foregoing aspects, features and advantages of the present technology will be further appreciated when considered with reference to the following description of preferred embodiments and accompanying drawings, wherein like reference numerals represent like elements. In describing the preferred embodiments of the technology illustrated in the appended drawings, specific terminology will be used for the sake of clarity. The invention, however, is not intended to be limited to the specific terms used, and it is to be understood that each specific term includes equivalents that operate in a similar manner to accomplish a similar purpose.

(11) According to one embodiment of the technology, a fast response electricity storage, or power storage system (PSS) can be provided to supply power to the power generation equipment of an electric hydraulic fracturing fleet when demand is high or in the event of a generator failure. The PSS system can include either solid state batteries or flow batteries. Solid state batteries can include, for example, electrochemical capacitors, lithium ion batteries, nickel-cadmium batteries, and sodium sulfur batteries. In addition, solid state batteries can charge or discharge based on electricity usage, and such charging and discharging can be paired with a software system, to monitor the state of the batteries and control the charging and discharging of the batteries. Flow batteries can, for example, include redox, iron-chromium, vanadium redox, and zinc-bromine batteries, and can be rechargeable batteries that store electricity directly in an electrolyte solution and respond quickly as needed. The flow batteries can also be paired with software, and the software associated with the both solid state and flow batteries can be designed to integrate with an operator's existing system so that monitoring and control can be integrated with other functions.

(12) FIG. 1A shows a hydraulic fracturing power system **100** according to an embodiment of the present technology. The hydraulic fracturing power system **100** includes a power source **110**, which can be, for example, a generator, and which can feed a first circuit breaker **120**. As shown, the hydraulic fracturing power system **100** can further include a PSS **130** that can feed a second circuit breaker **140**. In some embodiments, both the first circuit breaker **120** and the second circuit breaker **140** can be housed in the same switchgear housing **150**, or trailer. Both the first circuit breaker **120** and the second circuit breaker **140** can be connected to a common bus **160**, which in some embodiments can be a large copper bar used to share power evenly to downstream equipment from upstream generators.

(13) When the power source **110** is energized with both the first and second breakers **120**, **140** closed, hydraulic fracturing equipment **170** can be supplied power while the PSS **130** stores excess electricity. The hydraulic fracturing equipment can be hydraulic fracturing pumps, blenders data vans, wireline equipment, boost pumps, cranes, lighting, chemical trailers, etc. Once load requirements increase for the equipment **170**, the PSS **130** can release its stored power onto the common bus **160** in order to reduce the load on the power source **110**. The power source **110** and the PSS **130** can share the burden of supplying power during stages of high power demand until the end of the fracturing stage. Before the next fracturing stage begins, the PSS **130** can replenish stored electricity used previously until it is needed to discharge its power. This ability to recharge and discharge intermittently or continuously as needed ensures adequate power distribution to the system by the PSS **130** throughout an operation.

(14) Also shown in FIG. 1A are third circuit breaker **180** and fourth circuit breaker **190**. Each of the third and fourth circuit breakers **180, 190** can be electrically connected to equipment **170**. In the embodiment shown in FIG. 1A, each of the third and fourth circuit breakers **180, 190** are shown connected to pieces of equipment **170**, such as, for example, two hydraulic fracturing pumps. In practice, however, the present technology contemplates any appropriate ratio of circuit breakers to equipment, including connecting each circuit breaker to a single piece of equipment, or connecting each circuit breaker to more than two pieces of equipment.

(15) One advantage to the present technology is that it is a more efficient way of providing power at peak times than known systems, such as simply providing another generator on site. In addition, the entire PSS package can be much smaller than a second generator, thereby taking up less space on a pad. The storage system will also require significantly less rig up time due to having no fuel connections, crane lifts, or mechanical alignments.

(16) FIG. 1B is a schematic depiction of the PSS **130** of the embodiment of the hydraulic fracturing power system **100** of FIG. 1A. The PSS **130** can include a plurality of battery banks **131**, each connected to a common PSS bus **132** via an optional battery bank circuit breaker **133**. The common PSS bus **132** is also connected to a PSS circuit breaker **134** which is in turn electrically connected to circuit breaker **140** in the switchgear housing **150**.

(17) Each of the connections in the PSS **130**—between the battery banks **131** and battery bank circuit breakers **133**, the battery bank circuit breakers **133** and the common PSS bus **132**, the common PSS bus **132** and the PSS circuit breaker **134**, and the PSS circuit breaker **134** and the second circuit breaker **140**—are two way connections, as indicated by double headed arrows. This means that electricity flows in both directions between the various components. One advantage to this configuration is the ability of the battery banks **131** within the PSS **130** to constantly discharge and recharge as needed or allowed by the load demands of the system. Thus, when a heavy load is required, the PSS **130** can augment the power provided by power source **110** to help avoid overloading power source **110**. Conversely, when a light load is required, the PSS **130** can pull excess power from power source **110** to recharge battery banks **131**.

(18) Referring now to FIG. 2A, there is shown an alternate hydraulic fracturing power system **200** according to an alternate embodiment of the present technology, including a power source **210** and a PSS **230**. According to FIG. 2, the PSS **230** can be connected to the power source **210** in series before feeding power to a circuit breaker **215** in the switchgear housing **250**. Upon reaching full capacity, the PSS **230** can disconnect internal batteries from the power source **210**, thereby allowing it to bypass straight to the switchgear system.

(19) In the configuration shown in FIG. 2A, the circuit breaker **215** will then act as a feeder breaker for two additional circuit breakers **280, 290**. As shown, the circuit breaker **215** can be connected to circuit breakers **280, 290** via common bus **260**. Circuit breaker **215** can be rated for higher amperage than circuit breakers **280, 290**. Circuit breakers **280, 290** are in turn connected to hydraulic fracturing equipment **270**. Each of the additional circuit breakers **280, 290** can be electrically connected to equipment **270**. In the embodiment shown in FIG. 2A, each of the additional circuit breakers **280, 290** are shown connected to two pieces of equipment **270**, such as, for example, two hydraulic fracturing pumps. In practice, however, the present technology contemplates any appropriate ratio of circuit breakers to equipment, including connecting each circuit breaker to a single piece of equipment, or connecting each circuit breaker to more than two pieces of equipment.

(20) FIG. 2B is a schematic depiction of the PSS **230** of the embodiment of the hydraulic fracturing power system **200** of FIG. 2A. The PSS **230** can include a plurality of battery banks **231**, each connected to a common PSS bus **232** via a battery bank circuit breaker **233**. The common PSS bus **232** is also connected to an incoming PSS circuit breaker **235** and an outgoing PSS circuit breaker **236**. Outgoing PSS circuit breaker **236** is in turn electrically connected to circuit breaker **215** in the switchgear housing **250**.

(21) Many of the connections in the PSS **230**—between the battery banks **231** and battery bank circuit breakers **233**, and the battery bank circuit breakers **233** and the common PSS bus **232**—are two way connections, as indicated by double headed arrows. This means that electricity flows in both directions between the various components. One advantage to this configuration is the ability of the battery banks **231** within the PSS **230** to constantly discharge and recharge as needed. During a typical operation, power will discharge from the battery banks **231** to the circuit breaker **215** via the battery bank circuit breakers **233**, the common PSS bus **232**, and the outgoing PSS circuit breaker **236**. Simultaneously, or as needed, power from the power source will recharge the battery banks **231** via the incoming PSS circuit breaker **235**, the common bus **232**, and the battery bank circuit breakers **233**.

(22) As shown in FIG. 3A, in certain embodiments of the technology, the hydraulic fracturing power system **300A** can alternatively be powered by power transmission lines **305**, with the power source **310** and the PSS **330** providing parallel power to the switchgear **350**. In such an embodiment, the power source **310** and the PSS **330** can each be attached to circuit breakers within the switchgear housing, which are in turn connected to the hydraulic fracturing equipment **370**. This arrangement is similar to the embodiment shown in FIG. 1A, except that the power source **310** and the PSS **330** can be located at a remote location. The configuration of the circuit breakers within the switchgear housing **350** can be substantially similar to that of circuit breakers **120**, **140**, **180**, **190** in the embodiment shown in FIG. 1A. In addition, the PSS **330** can have a similar structure to that described above and shown in FIG. 1B.

(23) The arrangement shown in FIG. 3A, including the use of power transmission lines **305**, could be beneficial if, for example, space at a well site is restricted, and power generation has to be stationed some distance from the pad. In such an embodiment, cables can be sized properly due to distance, and additional protection can be installed for safety reasons, such as three phase reclosers **325** (small circuit breakers placed at distribution poles to clear faults on cables that are running long distances). In the embodiment of FIG. 3, the PSS **330** can be connected to the transmission lines for remote operations, but may still draw power from the power source **310**.

(24) FIG. 3B shows an embodiment of the hydraulic fracturing power system **300B** that shares characteristics of the embodiments of FIGS. 2A and 3A. That is, both the power source **310** and the PSS **330** are located at a remote location from the switchgear **350**, and they are connected to the switchgear **350** in series. One advantage to this embodiment is that it requires only one set of transmission lines **305** between the power source **310**/PSS **330** and the switchgear **350**. In this embodiment, the configuration of the circuit breakers within the switchgear housing **350** can be substantially similar to that of circuit breakers **215**, **280**, **290** in the embodiment shown in FIG. 2A. In addition, the PSS **330** can have a similar structure to that described above and shown in FIG. 2B.

(25) In yet another embodiment, shown in FIG. 4A, the hydraulic fracturing power system **400A** can include similar features to the embodiment shown in FIG. 3A, including a power source **410** and a PSS **430**. Moreover, the power source **410** is connected to the switchgear **450** via power transmission lines **405**, and the power transmission lines can include safety features, such as reclosers **425**. In the embodiment of FIG. 4A, the PSS **430** can also provide ancillary power. For example, if the power source **410** is a generator, and the generator shuts down during a fracturing stage, the PSS **430** can provide power to hydraulic fracturing equipment **470**, including pumps, in order to flush the well so that chemicals and sand previously being pumped through the well can be completely removed from the well.

(26) FIG. 4B shows an embodiment of the hydraulic fracturing power system **400B** that shares characteristics of the embodiments of FIGS. 2A and 4A. That is, the power source **410** is located at a remote location switchgear **350**, the PSS **430** is located at the well site, and the power source **410** and PSS **430** are connected to the switchgear **450** in series. One advantage to this embodiment is that the PSS **430** can provide power to the hydraulic fracturing equipment **470** even if the transmission lines **405** fail. Another advantage is that placing the PSS **430** at the wellsite allows for

the provision of power at the wellsite without any local emissions or appreciative noise. In this embodiment, the configuration of the circuit breakers within the switchgear housing **450** can be substantially similar to that of circuit breakers **215, 280, 290** in the embodiment shown in FIG. 2A. In addition, the PSS **430** can have a similar structure to that described above and shown in FIG. 2B.

(27) Another alternative embodiment of the present technology provides a hydraulic fracturing power system where the PSS can be used as black start for a power source that is a generator. Black starting is the process of supplying power to a generator that has been completely shut down to get it back up and running. Black start power can be used to power many different systems internal to a primary generator, including, for example, lighting, controls, blowers, cooling systems, lube pumps, oil pumps, starting motors, etc, until the generator is up and running and can provide its own power for these ancillary systems. Diesel generators can usually do this with battery power, but turbine generators require a larger power source, especially if gas compressors need to be operating before the engine can be fired. The configuration of the PSS relative to the switchgear and equipment in such a case can be similar to the embodiments shown in FIGS. 1A-4B. If enough power is stored in the batteries, the PSS system could support black starting operations without the need for a smaller standby generator to act as the black start power source. However, it could also utilize an external power source, such as solar panels, to recharge the storage system.

(28) Use of the PSS in hydraulic fracturing power system of the present technology provides numerous advantages over known systems, including load leveling, frequency regulation, power quality control, emergency power, black start power, load bank capabilities, equipment reduction, reduced maintenance, and a simplified fuel supply. Each of these features is discussed in detail herein below.

(29) First, with regard to load leveling, the PSS of the present technology has the ability to store electricity in times of low demand, and then to release that electricity in times of high power demand. As applied to electric powered hydraulic fracturing, stages that require relatively less load can provide a time for the PSS to charge up, or store electricity. In addition, the PSS can charge between stages or at the beginning of stages before full pump rate is achieved. Thereafter, power can be released in the stages of higher load requirements. This helps in increasing the lifespan of a power generating asset by decreasing its workload.

(30) With regard to frequency regulation, the PSS can charge and discharge in response to an increase or decrease of microgrid frequency to maintain stored electricity within prescribed limits. This increases grid stability. In other words, the PSS can ramp up or down a generating asset in order to synchronize the generator with microgrid operation.

(31) With regard to power quality control, the PSS can protect downstream loads such as sensitive electronic equipment and microprocessor based controls against short-duration disturbances in the microgrid that might affect their operation.

(32) With regard to emergency power, in the event of a generator failure (due to, for example, a mechanical fault, electric fault, or due to a fuel supply loss), the PSS can provide sufficient electric power to flush the wellbore. This feature can prevent a “screen out” where the loss of fluid velocity causes the proppant in the hydraulic fracturing fluid or slurry to drop out and settle in the wellbore. Such a screen out can plug off the perforations and cause several days of downtime to clear. A screen out is a major concern in hydraulic fracturing and is considered a failure. The PSS can allow an electric hydraulic fracturing fleet to properly flush the well by being able to power the electric blender as well as sufficient hydraulic fracturing pumps to displace the proppant-laden slurry completely into the formation without generator power.

(33) With regard to black start power, normally a small generator can be used to provide power to ancillary systems such as heaters, blowers, sensors, lighting, programmable logic controllers, electric over hydraulic systems, and electric over air systems for the larger generators. Such a generator can also be used to power the starters for these larger generators, which are often electric starters with a variable frequency drive or soft starter, or can be hydraulic starters with electric

motors powering the hydraulic pumps. If the PSS is properly charged, it can replace the black start generator to allow the larger generators (often turbines) to start from a black out condition.

(34) With regard to load bank capabilities, the PSS can be used to test and verify generator performance during commissioning or after mobilization. It can also work for load rejections, to dissipate power during sudden shut downs, such as if the wellhead exceeds the maximum pressure and every frac pump needs to shut down simultaneously without warning.

(35) With regard to equipment reduction, using an electricity storage system can allow electric fracturing operations to eliminate or reduce the use of a black start generator or supplemental generator, or a standby generator. Many times more than one large turbine generator is desired to provide power during peak demand during a hydraulic fracturing stage. Other times, a secondary generator can be held electrically isolated in standby in the event of a primary generator failure. Such secondary turbines can be replaced by the PSS, resulting in lower noise levels, less equipment on a pad, and faster mobilization times between well sites.

(36) With regard to the reduced maintenance requirements, in some embodiments the PSS can be comprised of a solid state battery bank having very few moving parts. Thus, the PSS will require less maintenance than a generator utilizing a turbine or reciprocating engine.

(37) With regard to the simplified fuel supply, in embodiments where the PSS is replacing a secondary or standby generator, the PSS will not require any fuel supply as it can be energized by a power grid. Therefore, any fuel connections for liquid or gas fuel can be removed from the system. This allows for a reduction in the number of connections and manifolds, as well as a reduction in the fuel volumes required during peak demand. In embodiments where the PSS replaces, for example, one of two turbines, all of the fuel equipment, hoses, and manifolding can be greatly reduced and simplified.

(38) Although the technology herein has been described with reference to particular embodiments, it is to be understood that these embodiments are merely illustrative of the principles and applications of the present technology. It is therefore to be understood that numerous modifications may be made to the illustrative embodiments and that other arrangements may be devised without departing from the spirit and scope of the present technology as defined by the appended claims.

Claims

1. A hydraulic fracturing power system, comprising: a power source; a power storage system; an electric powered hydraulic fracturing pump in selective electrical communication with the power storage system; and a circuit breaker between the power storage system and the electric powered hydraulic fracturing pump, the circuit breaker having an open position that opens an electric circuit between the electric powered hydraulic fracturing pump and the power storage system, and a closed position that closes the electric circuit, the circuit breaker varying between the open position and the closed position as required to power the electric powered hydraulic fracturing pump while the electric powered hydraulic fracturing pumps are in place at a wellsite, wherein the circuit breaker comprises a first circuit breaker, and the hydraulic fracturing power system further comprising: a power source in selective electric al communication with the electric powered hydraulic fracturing pump, and a second circuit breaker, wherein the second circuit breaker is electrically connected to the power source, and wherein the first circuit breaker is electrically connected to the power storage system.
2. The system of claim 1, wherein the power storage system is at least one solid state battery selected from the group consisting of electrochemical capacitors, lithium ion batteries, nickel-cadmium batteries, and sodium sulfur batteries.
3. The system of claim 2, wherein the power storage system is at least one flow battery selected from the group consisting of redox batteries, iron-chromium batteries, vanadium redox batteries, and zinc-bromine batteries.

4. The system of claim 3, wherein the at least one solid state battery or the at least one flow battery is rechargeable.
5. The system of claim 1, wherein each of the first circuit breaker and the second circuit breaker is electrically connected to the electric powered hydraulic fracturing pump via a common bus.
6. The system of claim 1, wherein the circuit breaker comprises a first circuit breaker, and further comprising: a power source in selective electrical communication with the electric powered hydraulic fracturing pump, wherein both the power source and the power storage system are electrically connected to the first circuit breaker.
7. The system of claim 1, wherein the power storage system is electrically connected to the circuit breaker via a power line.
8. The system of claim 1, wherein the power storage system is mounted to a trailer.
9. The system of claim 1, wherein the circuit breaker is substantially enclosed in a switchgear housing.
10. A system for powering an electric hydraulic fracturing pump, comprising: a power storage system having; an electric powered hydraulic fracturing pump in selective electrical communication with the power storage system; and a circuit breaker between the power storage system and the electric powered hydraulic fracturing pump, the circuit breaker configured to facilitate or prevent electrical communication between the power storage system and the electric powered hydraulic fracturing pump while the electric powered hydraulic fracturing pumps are in place at a wellsite, wherein the circuit breaker comprises a first circuit breaker, and further comprising: a power source, and a second circuit breaker, wherein the second circuit breaker is electrically connected to the power source, and the second circuit breaker is electrically connected to the power storage system.
11. The system of claim 10, further comprising: a power source, wherein both the power source and the power storage system are electrically connected to the circuit breaker, and the circuit breaker facilitates or prevents communication between the power storage system, the power source, and the electric powered hydraulic fracturing pump as required to power the electric powered hydraulic fracturing pump and maintain a charge in the power storage system.
12. The system of claim 11, wherein at least one of the power source and the power storage system are electrically connected to the first circuit breaker or the second circuit breaker via a power line.
13. The system of claim 10, wherein the circuit breaker is substantially enclosed in a switchgear housing.
14. The system of claim 13, further comprising: software in communication with the power storage system, the software configured to monitor a state of the power storage system and to integrate control of the power storage system with other features of the system for powering electric hydraulic fracturing equipment.
15. The system of claim 10, wherein the power storage system is rechargeable.
16. The system of claim 10, wherein the power source is electrically connected to the first circuit breaker via a power line, and wherein the power storage system is located adjacent a switchgear housing and electrically coupled directly to the switchgear housing without a power line.
17. A system for powering an electric hydraulic fracturing pump, comprising: a power storage system having: an electric powered hydraulic fracturing pump in selective electrical communication with the power storage system; and a circuit breaker between the power storage system and the electric powered hydraulic fracturing pump, the circuit breaker configured to facilitate or prevent electrical communication between the power storage system and the electric powered hydraulic fracturing pump while the electric powered hydraulic fracturing pumps are in place at a wellsite, wherein the power storage system is at least one solid state battery selected from the group consisting of electrochemical capacitors, lithium ion batteries, nickel-cadmium batteries, and sodium sulfur batteries.
18. The system of claim 17, wherein the power storage system is at least one flow battery selected

from the group consisting of redox batteries, iron-chromium batteries, vanadium redox batteries, and zinc-bromine batteries.
